

Electrical Engineering Line-up

ASI Job #: (x1) 8855-200
Customer: Adient Riverside, MO
Package Release: ASAP
Hardware Due: ASAP
Panel Complete: 09/18/26
Runoff Date: 10/05/26
Fixture Description: P736 1st Row FSC Vent Bag Instillation Fixture
PLC Enclosure: 24 x 30 x 8 Enclosure RITTALL; Part # EMRA3WM8017549A015 SUB Plate incl.
PLC: Allen-Bradley Compact Logix L16 – Part # 1769-L16ER-BB1B
HMI Enclosure: 10" RITTALL w/ Swivel Arm EMRA3AX6315450A013 & EMRA3CP6206380A005
HMI: Allen-Bradley PanelView 1000 Part # 2711PT10C21D8S
PLC IP: XXX.XXX.X.XX
HMI IP: XXX.XXX.X.XX
Cycle Start Style:



Cycle Time: 52 seconds

Fixture outline:

Automated Solutions will design and manufacture (x1) 1st row FSB Vent bag and Heat pad installation fixture for the P736 program. The station will feature a compression cylinder, Keyence VS Vision system, Keyence IV4 Vision system, Main Part present sensor, LH vs RH present sensors, and Keyence photoelectric sensors.

Machine Home Condition:

- Compression cylinder retracted and all sensors off besides cylinder cap sensor.

Assembly Sequence of Operations:

1. Operator scans build ticket and will obtain all correct parts
2. Operator obtains vent bag (If applicable)
3. Operator scans barcode on vent bag, peels tape, and loads vent bag onto the fixture using the alignment pins.
4. Operator loads Foam to nest. It is aligned with pins on the fixture and nest. If proper foam is loaded the sequence will advance to next step.
5. Operator obtains heater mat (If applicable)
6. Operator will apply heater mat and scan barcode on heater mat (via MAS) to assure presence and correct part
7. Operator swipes duo opto touch to activate compression cylinder.
8. Timer starts at full compressed state.
 - a. Compression force and time will be a minimum of 1N/cm² and pressed for a minimum 2 seconds.

9. Operator is notified through stack light and swipes opto to release compression.
10. After compression, Camera will confirm correct placement and no touching wires and then store image and pass data.
11. Operator will remove part from the fixture. Fixture will reset sequence because part in place was lost.
12. In event of Camera failure, operator can adjust heat pad then hit switches for retry of vision
13. Operator removes finished foam and places into WIP

Major Components on Machine:

- (x1) Allen-Bradley Compact Logix L16 – Part # 1769-L16ER-BB1B
- (x1) Allen-Bradley PanelView 1000 Part # 2711PT10C21D8S
- (x1) Dual Ethernet programming port ~ Mencom Part # MEN-DP1-2RJ45-16LS
 - Mount on side of enclosure
- (x1) Stack Light – Banner Pt. # TL50GRBAQ
 - (x1) Cable for stack light - IFM Part # EVC071
- (x1) E-stop Button – Banner Part # SSA-EB1PLYR-12ECQ8
 - (x1) IFM QD Cable for E-stop – Part # E11807
- (x1) E-Stop Relay ~ Banner Part #ES-FA-9AA
- (x2) Banner Opto Touch Button ~ Part #STBVR81Q6 (64192)
 - (x2) IFM QD Cable for opto touch ~ Part #EVC071
- (x1) Cycle Start Relay ~ Banner Part #AT-FM-10K (60698)
- (x1) Allen-Bradley Green System Start Button ~ Part #800T-QT24G
 - Mounted on panel door under HMI
- (x1) Allen-Bradley 2 POS. Spring Return Key Switch - Pt. # 800T-H42A
 - Mounted on panel door under HMI
- (x1) Hirschmann 8 PORT E-NET Switch Pt. # 942132017
- (x1) 2' Operator light and cord – Part # V087122100, V087032100
- (x8) Keyence photoelectric sensor – Part # PR-FB15C3PL
 - (x8) IFM 5m M8 3 Pin Cables – Part # EVC142
- (x1) Keyence VS Smart Camera (Supplied by Customer Except Swivellink Components)
 - (x1) VS smart camera – Part # VS-L1500CX
 - (x1) M12 to RJ45 10M Ethernet Cable – Part # OP-88837
 - (x1) Led lighting smart ring light – Part # CA-DEW10X
 - (x1) Polarizing filter for smart light – Part # OP-88807
 - (x1) 10m M12 to flying lead power and I/O cable – Part # OP-88812
 - (x1) VS mounting plate – Part # VSPLATE
 - (x1) VS AI card – Part # VS-AD1
 - (x2) Standard Ball Base ~ Swivellink Pt. AFSB-1
 - (x2) Standard Ball Knuckle ~ Swivellink Pt. AFSB-2
 - (x1) Standard Link 4" ~ Swivellink Pt. AFSB-3-4
- (x1) Bar code scanner: Equipment to be supplied & integrated by customer.
- (x1) MIRA (SCADA): Equipment to be supplied & integrated by customer.

Mechanical Components (On MBOM):

- (x1) Custom aluminum extrusion frame
- (x4) Linak Ergo jacks & controller
- (x1) SMC Manifold sub assembly required includes all below
 - (x1) 4-Station manifold – Part # VV5QC41-04N7TFD0
 - (x1) 3 position double solenoid valve – Part # VQC4501-51 **Valve #1**
 - (x1) Compression Cylinder
 - (x1) 2 position single solenoid valve – Part #VQ4101-5W **Valve #2**
 - (x1) Compression Cylinder Break
 - (x1) Sandwich Regulator – Part # ARBQ4000-N0-X
 - (x2) Blank plate – Part # VVQ4000-10A-1
 - (x1) Compression Fine Lock Cylinder – Part # CDLA2F100TN-600-E

INPUTS

- (1) Control Power on
- (1) Key switch Manual mode
- (3) Adv/Ret Compression Cylinder Switch
- (1) Cycle Start Relay
- (1) Pressure Switch
- (8) Keyence Photoelectric Sensors

15 Inputs

OUTPUTS

- (1) Control Power on button light
- (1) Advance Compression Cylinder Break
- (2) Adv/Ret Compression Cylinder Valve
- (5) Adv/Ret Spare Valves
- (4) Stack Light
- (1) Overhead Light Relay

14 Outputs

Notes/Comments:

- Drawings files can be located on the server inside the job folder
- ASI standard hardware & controls
- Single operator station
- Leave valves energized in their work and home states

COMPRESSION
CYLINDER

ADV - I: 2/0
RET - I: 2/1
MID - I: 2/2
ADV SOL.A - O: 3/1
RET SOL.B - O: 3/2
BRAKE - O: 3/0

BANNER STACK
LIGHT

Green O: 1/13
Red O: 1/14
Blue O: 1/12
Audible O: 1/15

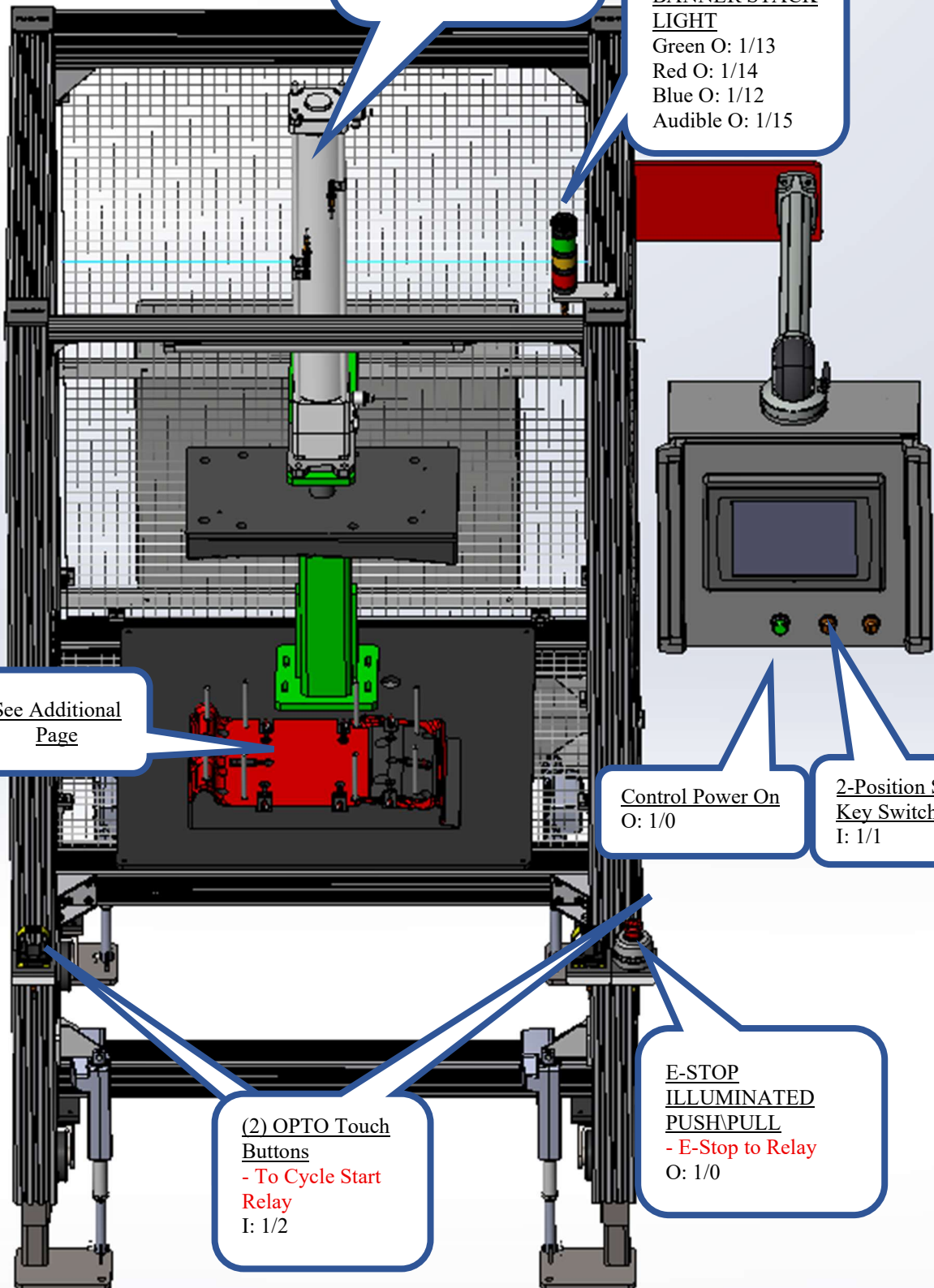
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Control Power On
O: 1/0

2-Position Spring
Key Switch
I: 1/1

(2) OPTO Touch
Buttons
- To Cycle Start
Relay
I: 1/2

E-STOP
ILLUMINATED
PUSH/PULL
- E-Stop to Relay
O: 1/0



Photoelectric
Sensor (1)
I: 1/8

Photoelectric
Sensor (2)
I: 1/9

Photoelectric
Sensor (3)
I: 1/10

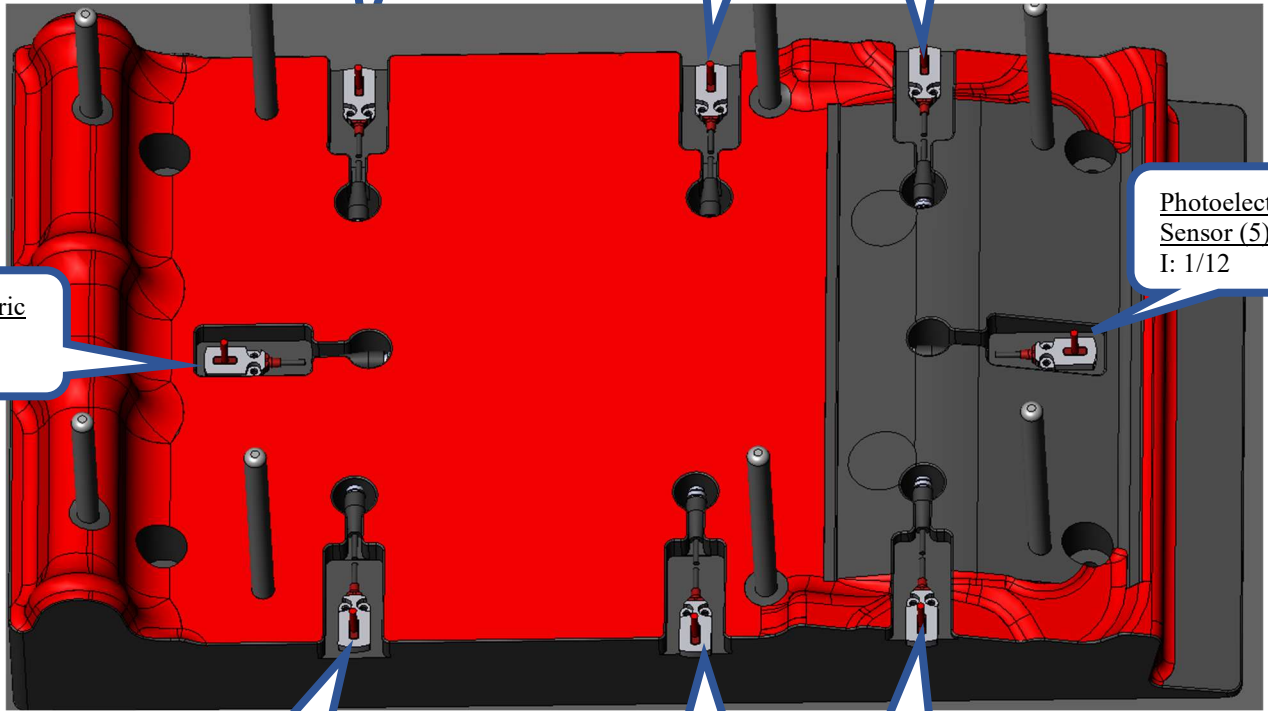
Photoelectric
Sensor (4)
I: 1/11

Photoelectric
Sensor (5)
I: 1/12

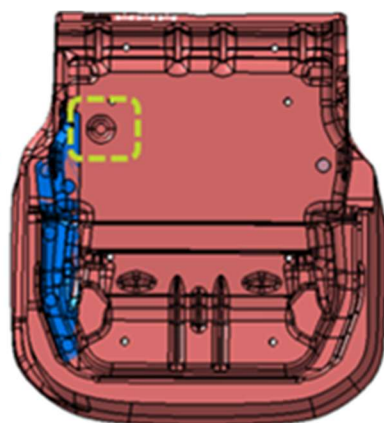
Photoelectric
Sensor (6)
I: 1/13

Photoelectric
Sensor (7)
I: 1/14

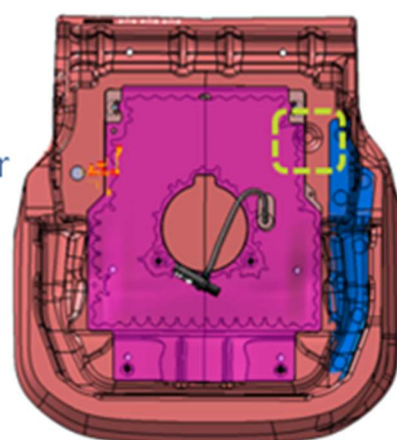
Photoelectric
Sensor (8)
I: 1/15



Driver



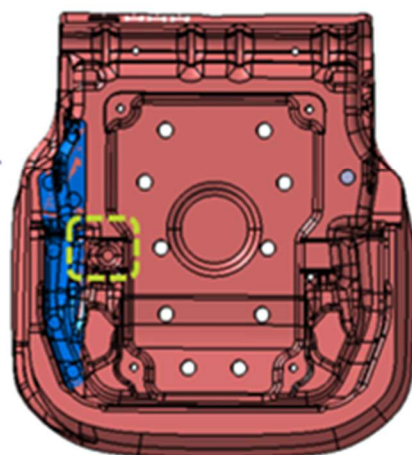
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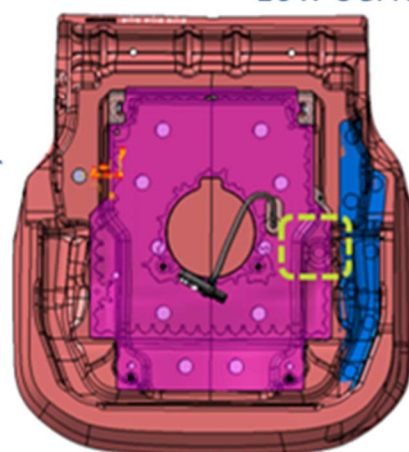
Low Series

Low Series

Driver



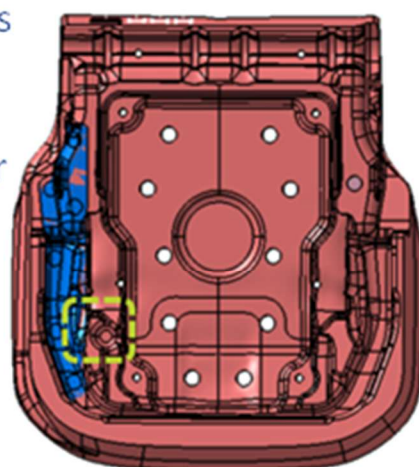
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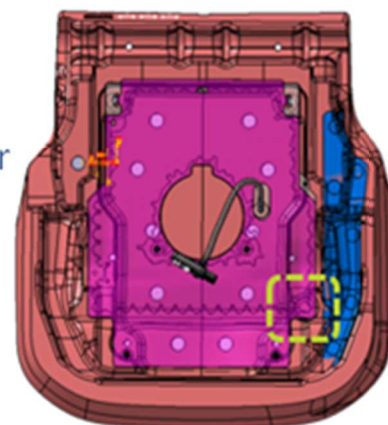
Mid Series

Mid Series

Driver



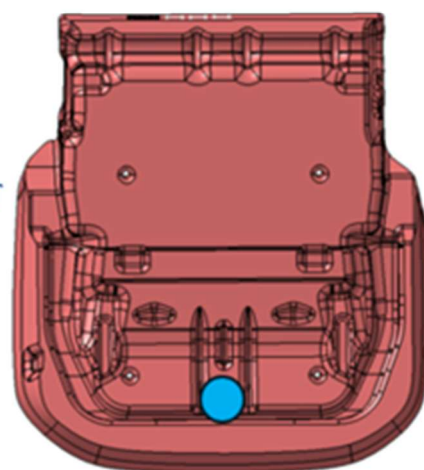
Passenger



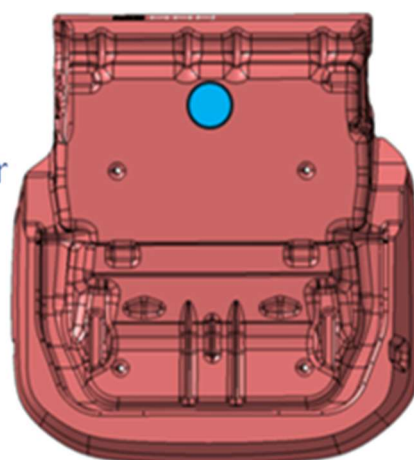
High Series

High Series

Driver



Passenger



Police Series

Police Series